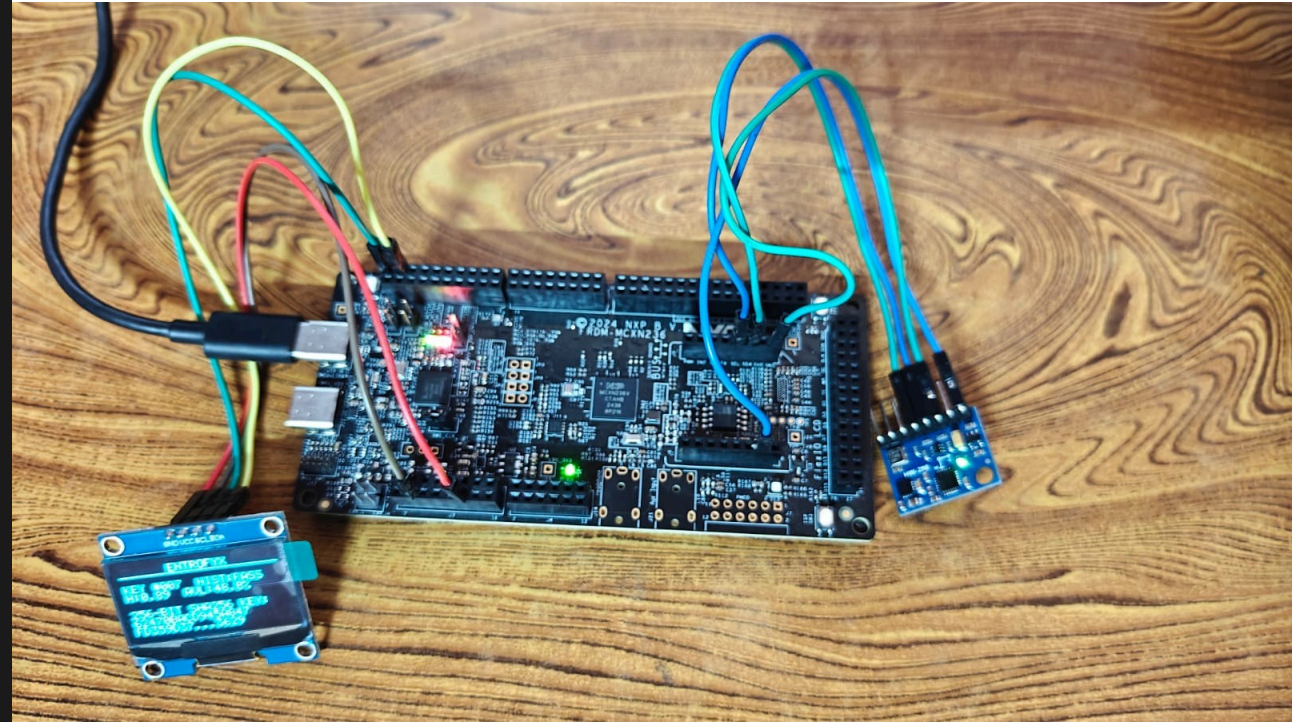


ENTROPYX ·

EntropyX: A physical true random number generator

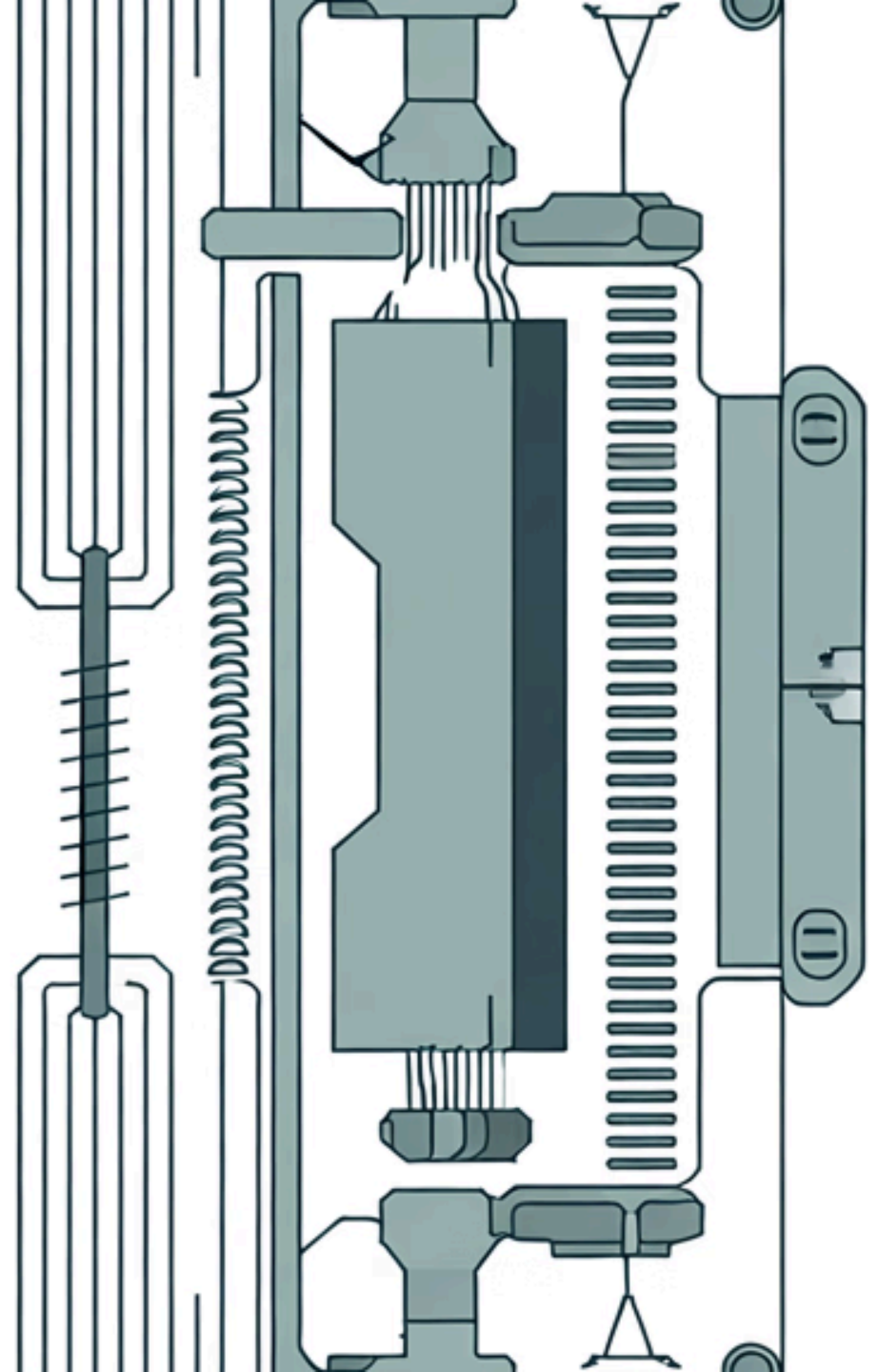
Inspectable physical noise, NIST SP 800-90B
qualification, and Toeplitz extraction

L K Prathika



*“Anyone who considers arithmetical methods of
producing random digits is, of course, in a state of sin.”*

— John von Neumann (1951).



Why measure physical entropy at the source?

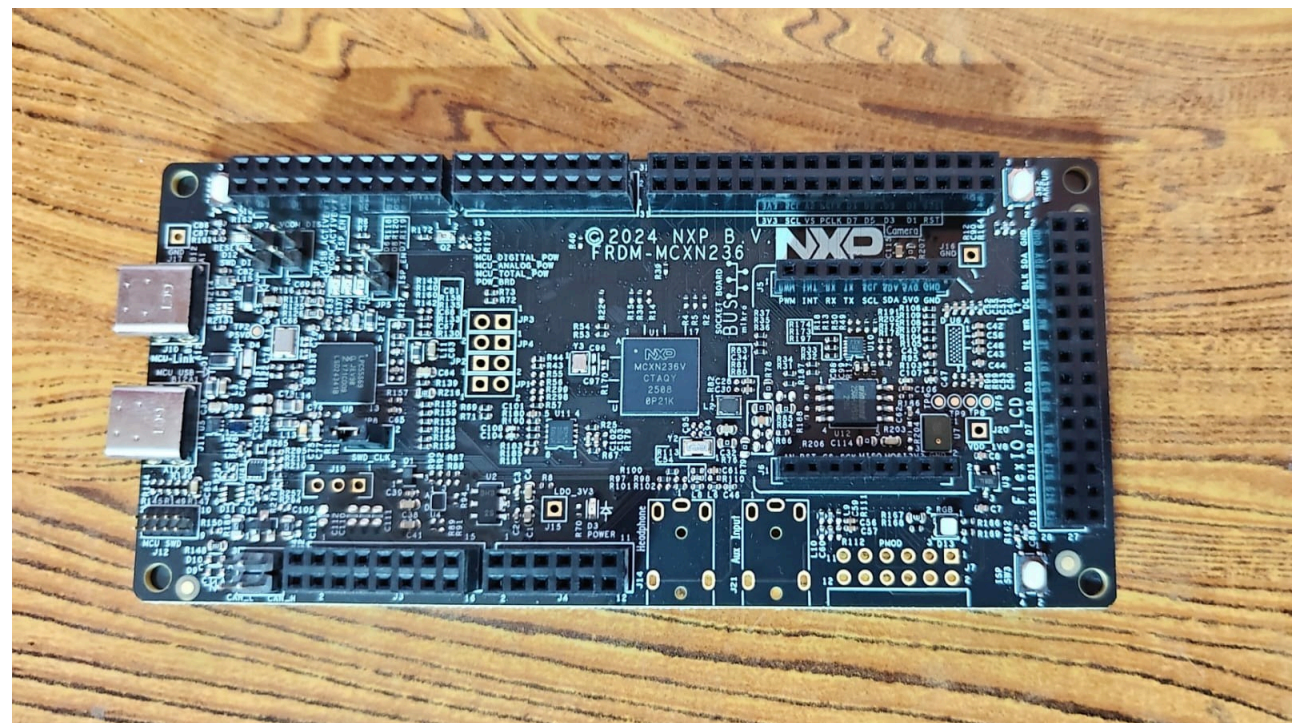
EntropyX replaces algorithmic PRNG dependence with inspectable thermodynamic and physical noise.

Context

Random values support secure keys, device identity, and session setup. Measuring the source makes bias, drift, and failure visible before conditioning.

What this project explores

The project combines sensor measurements, timing variation, on-chip tests, and a live display so the path from physical noise to output is traceable.



04 · PROBLEM STATEMENT

The prototype responds to two practical problems.

Predictability

Deterministic generators can become predictable after seed compromise; physical noise adds an independently observed source.

Limited visibility

Closed appliances can hide the evidence needed to judge live entropy quality.

The design challenge is to expose the source, the decision, and the output in one path.

INSPECT

See the source

Measure physical variation before it becomes key material.

DECIDE

Show the gate

Make qualification an explicit accept-or-reject step.

OUTPUT

Trace the result

Connect accepted input to the conditioned output.

05 · OBJECTIVES & NOVELTY

The novelty is the combination of physical sensing, on-chip qualification, extraction, and live evidence in one low-cost prototype.

01

Harvest

Collect physical and timing variation.

02

Qualify

Estimate and test entropy on-chip.

03

Condition

Turn accepted samples into 256-bit output.

04

Show

Make the decision visible on the device.

06 · HARDWARE ARCHITECTURE & COMPONENTS

The prototype keeps each stage inspectable, from physical input to conditioned key.

SENSE

10-DOF MEMS
cluster

GY-87 with MPU6050
accelerometer/gyroscop
e and QMC6308
magnetometer.

MIX

11 sampled
channels

Physical channels are
combined with timing
variation from the MCU.

TIME

DWT->CYCCNT at
150 MHz

Cycle-counter variation
supplements physical
sampling.

DECIDE

NXP FRDM-
MCXN236

Cortex-M33 runs
qualification and
extraction.

Sensor inputs → conditioning → NIST checks → Toeplitz hashing → OLED / dashboard.

Two buses keep sensing and display inspectable.

The low-cost prototype separates sensor traffic from the OLED path while the Cortex-M33 coordinates both.

I²C BUS 1 · SENSE

GY-87

MPU6050 accelerometer/gyroscope + QMC6308 magnetometer.

Sampled into the 11-channel physical-noise path.

I²C BUS 2 · DISPLAY

OLED + status

Shows entropy estimate, qualification result, key output, and avalanche evidence.

NXP FRDM-MCXM236 Cortex-M33 remains the decision point.

PROTOTYPE BOM

₹3,735

Low-cost, inspectable embedded hardware.

COMPARISON

₹55,000–₹2,00,000

Typical HSM price range cited for the comparison.

08 · PHYSICAL NOISE HARVESTING

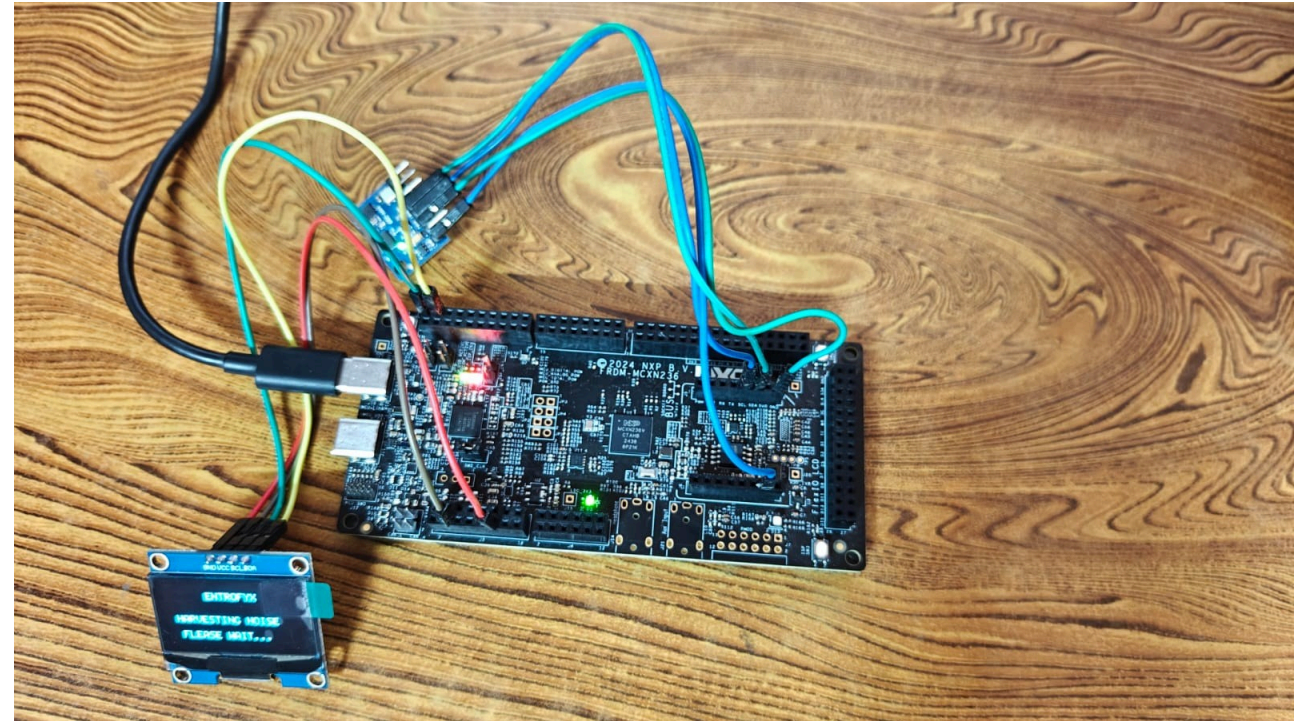
The sensor stack combines physical motion and timing variation before qualification.

- Johnson–Nyquist thermal agitation, Brownian motion, and CMOS clock jitter contribute distinct mechanisms.
- GY-87 channels provide the physical inputs: MPU6050 accelerometer/gyroscope and QMC6308 magnetometer.
- DWT->CYCCNT at 150 MHz adds a timing signal across the 11 sampled channels.

$$\Delta S(t) = S(t) - S(t - 1)$$

$$H(z) = 1 - z^{-1}$$

The first-order difference rejects static behavior before mixing.



Reject static behavior before extraction

Conditioning separates useful variation from samples that should not enter the key path.

FILTER

Difference

Remove static structure with the first-order filter $H(z)=1-z^{-1}$.

MIX

11 channels + timing

$\text{mix} = \Delta S(t) \oplus (\text{DWT} \rightarrow \text{CYCCNT} \ \& \ 0xFF)$

HAND OFF

1,024-bit blocks

Only conditioned blocks move to the NIST gatekeeper.

Accept only qualified blocks

Statistical checks run before any block is promoted to key material.

SP 800-90B

$$H_{\infty} = -\log_2(\max_x P[X = x])$$

Threshold: $H_{\infty} \geq 0.500$ bits/bit.

The result is a visible gate: accept the block, or reject it.

SP 800-22

Monobit $p \geq 0.01$ · Runs $p \geq 0.01$ · Serial χ^2 $p \geq 0.01$.

11 · TOEPLITZ EXTRACTION & SECURITY

Universal hashing turns accepted physical samples into a fixed 256-bit output.

Toeplitz hashing

$$y = T \cdot x \pmod{2}$$

$$T \in \mathbb{F}_2^{256 \times 1024}$$

A 1,024-bit qualified block is mapped to 256 bits with an explicit universal hash.

Assurance bound

$$\Delta \leq 2^{-333.8} \ll 2^{-64}$$

Using $k=921.6$ bits of source min-entropy and $m=256$ output bits.

FAIL-CLOSED PATH

Sense

↓ qualify

Accept or reject

↓ hash accepted blocks

256-bit output

12 · THREAT MODEL & PHYSICAL DEFENSES

The design treats a static, biased, or compromised source as a reason to stop and inspect.

Reject

Discard failed blocks before extraction; hold the system closed on bias or test failure.

Defend

Magnetic injection: compare magnetometer behavior with the other channels and reject anomalous blocks.

Bound

Cryogenic cooling lowers thermal variance; acoustic resonance can create periodic bias. Monitor drift, use qualification, and stop on failed thresholds.

13 · HARDWARE BENCHMARKS & EMPIRICAL RESULTS

The prototype surfaces the output key alongside entropy quality, NIST status, and avalanche behavior.

ENTROPY	AVALANCHE	THROUGHPUT
0.932±0.041	52.1%	2.78
Min-entropy, bits/bit.	±2.8%	s/key
	μ=50.0% ·	138.6 mW
	σ=3.125%.	at the
		prototype
		operating
		point.

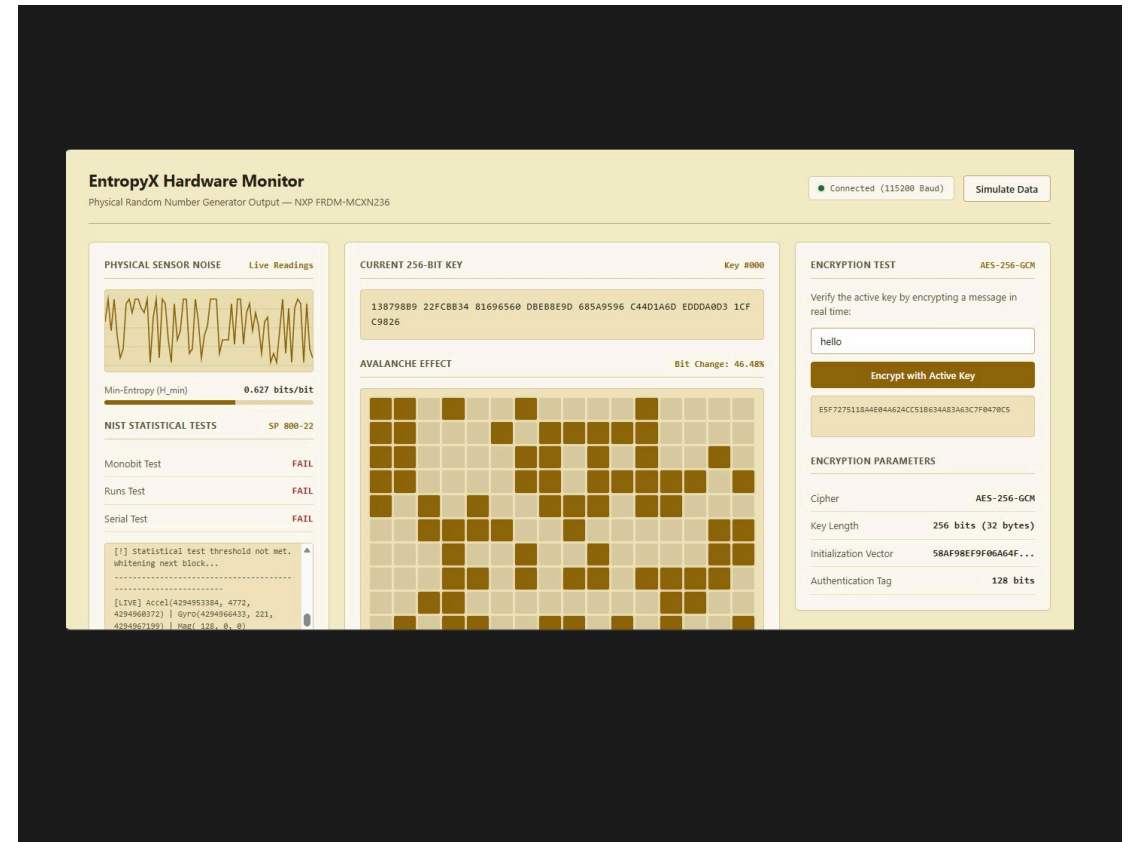


The dashboard

A live display links the generated key to the evidence that qualified it.

Shown live

- Current key output
- Min-entropy estimate
- NIST pass or fail
- Avalanche indicator
- Environmental and qualification status



15 · APPLICATIONS

EntropyX is a prototype, but its inspectable path points to several practical uses.

Embedded security

Device identity, session keys, and hardware-backed demonstrations.

Compared with software

Physical noise is observable; algorithmic state alone is not.

Inspectability

Low-cost sensing, qualification, and display make the path from source to output easier to inspect.

Scope: these are potential use cases, not a claim that the prototype is production-ready.

EntropyX proves that provable physical randomness can be harvested transparently and affordably.

1. ENGINEERING GOALS ACHIEVED

- Multi-modal transduction works: mechanical Brownian motion, Johnson-Nyquist thermal agitation, and geomagnetic micro-perturbations are harvested simultaneously.
- Deterministic offsets eliminated: the first-order discrete difference filter $H(z) = 1 - z^{-1}$ and Von Neumann de-biasing remove static gravitational (1G) and ambient magnetic baselines.
- Autonomous gatekeeping confirmed: real-time NIST SP 800-90B and SP 800-22 self-auditing runs inside bare-metal microcontroller SRAM and rejects biased blocks without external PC intervention.

3. LIMITATIONS & ENGINEERING TRADE-OFFS

- Throughput: about 92 bits/second of certified physical entropy; optimized for root key derivation, cold-storage seed generation, and session initialization rather than continuous bulk data streaming.
- Bandwidth limit: 1 key / 2.78 seconds.
- At extreme low temperatures ($T < -40^{\circ}\text{C}$), thermal noise variance drops, requiring longer pooling times to satisfy the min-entropy threshold.
- Multi-modal dependency requires balanced bus capacitance tuning.

2. QUANTITATIVE BENCHMARKS

- Min-entropy: 0.932 ± 0.041 bits/bit; NIST passing threshold $H_{\infty} \geq 0.500$ bits/bit.
- Statistical distance: $\Delta \leq 2^{-333.8}$ versus $\Delta \leq 2^{-64}$.
- Avalanche mean: $52.1\% \pm 2.8\%$; $\mu = 50.0\%$; $\sigma = 3.125\%$.
- One certified 256-bit key every 2.78 seconds at 138.6 mW.

4. FUTURE ROADMAP

- Custom 4-layer surface-mount PCB with a ground-shielded Faraday enclosure.
- Post-Quantum Cryptography integration with NIST-standardized algorithms (ML-KEM / CRYSTALS-Kyber and ML-DSA / CRYSTALS-Dilithium).
- Automated USB CCID token firmware: a standard driverless PKCS#11 cryptographic token.
- Environmental chamber stress testing (-20°C to $+70^{\circ}\text{C}$ thermal drift analysis).



FINAL TAKEAWAY

“True security cannot rely on deterministic software algorithms. EntropyX proves that provable, NIST-certified physical randomness can be harvested transparently and affordably in open embedded hardware.”

Thank you.

Questions?